

Support Vector Machines

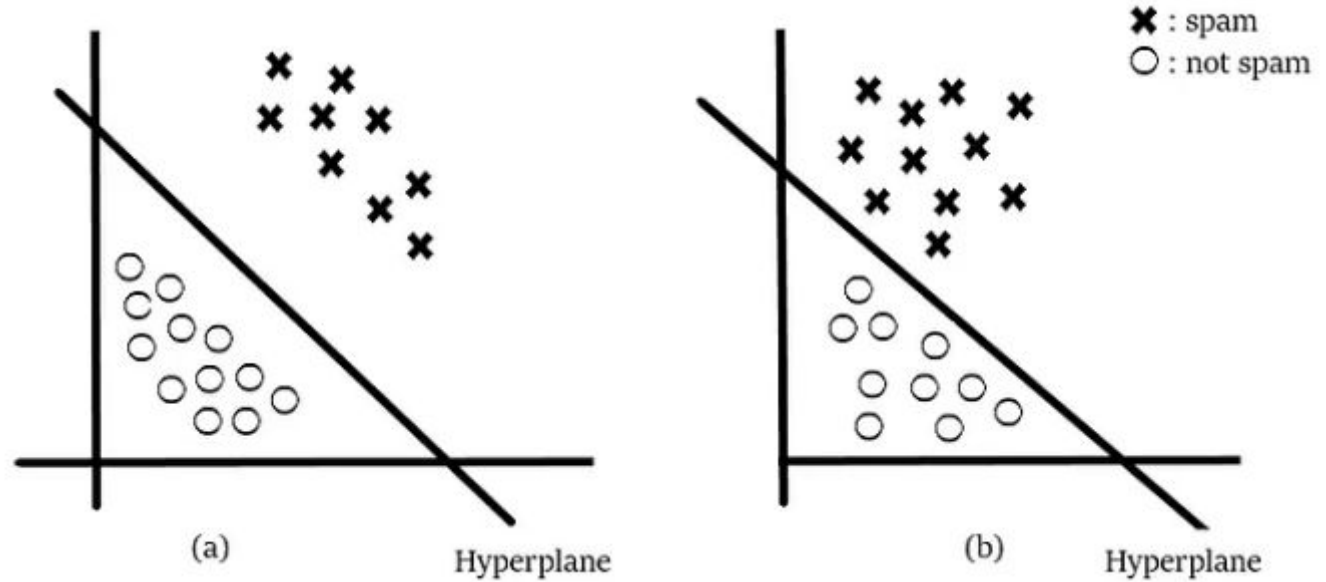
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Support Vector Machine (SVM)

- A powerful supervised learning algorithm for classification and regression
- **Key Idea:** Find the best separating hyperplane between different classes.
- **Hyperplane:** an $(n-1)$ dimensional plane on an n dimensional space

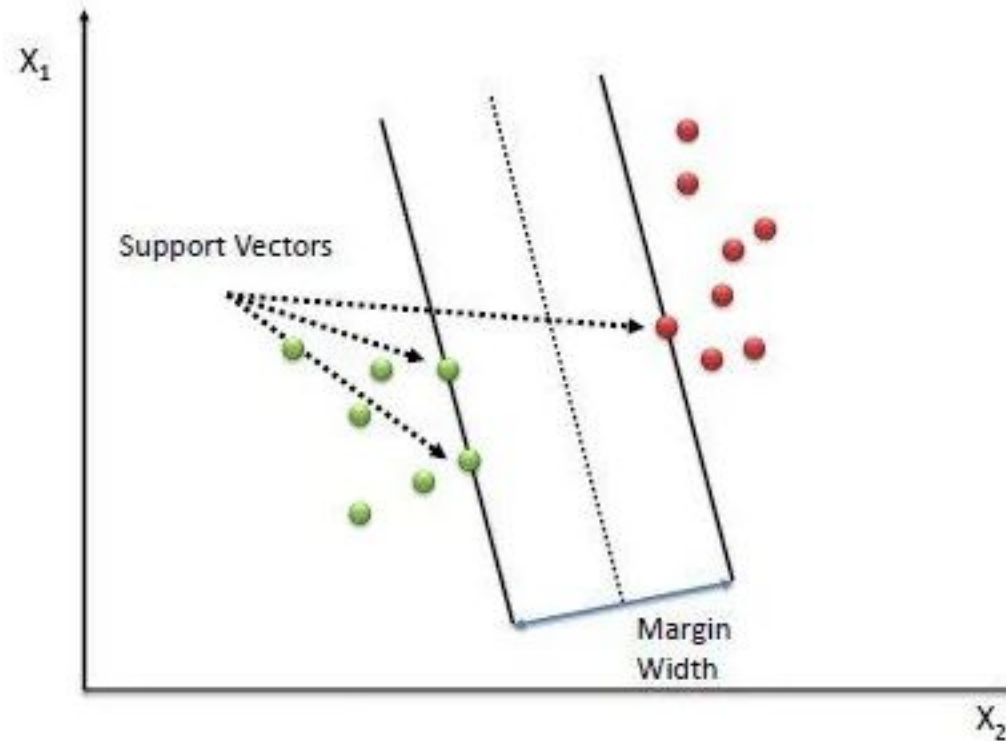
SVM Intuition

- The idea of coming up with an Optimal hyperplane that maximizes the margin between classes



Support Vector

- The points closest to the hyperplane are called as the **support vector points**
- The distance of the vectors from the hyperplane are called the **margins**



Hyperplane

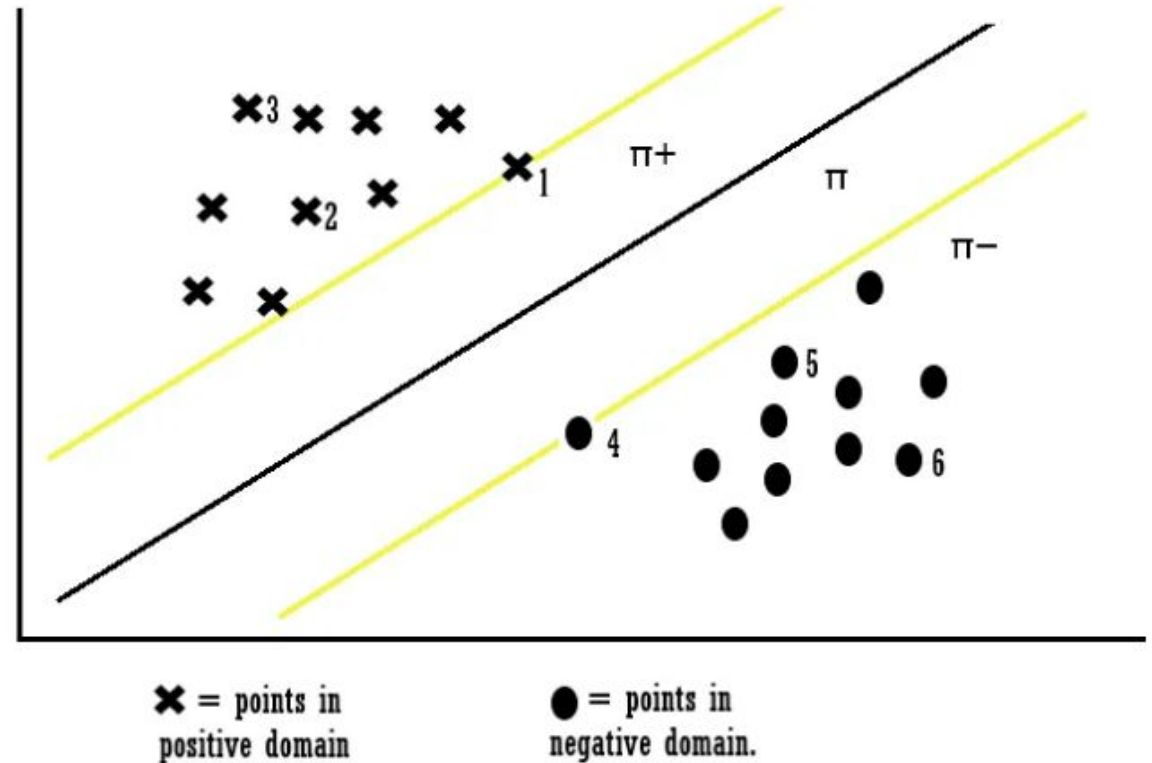
- We are given a training dataset of n points of the form $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$
- Any hyperplane can be written as the set of points $\mathbf{x}$ satisfying

$$w^T x + b = 0$$

$$b + w_1 x_1 + w_2 x_2 + \dots + w_n x_n = b + \sum_{i=1}^n w_i x_i$$

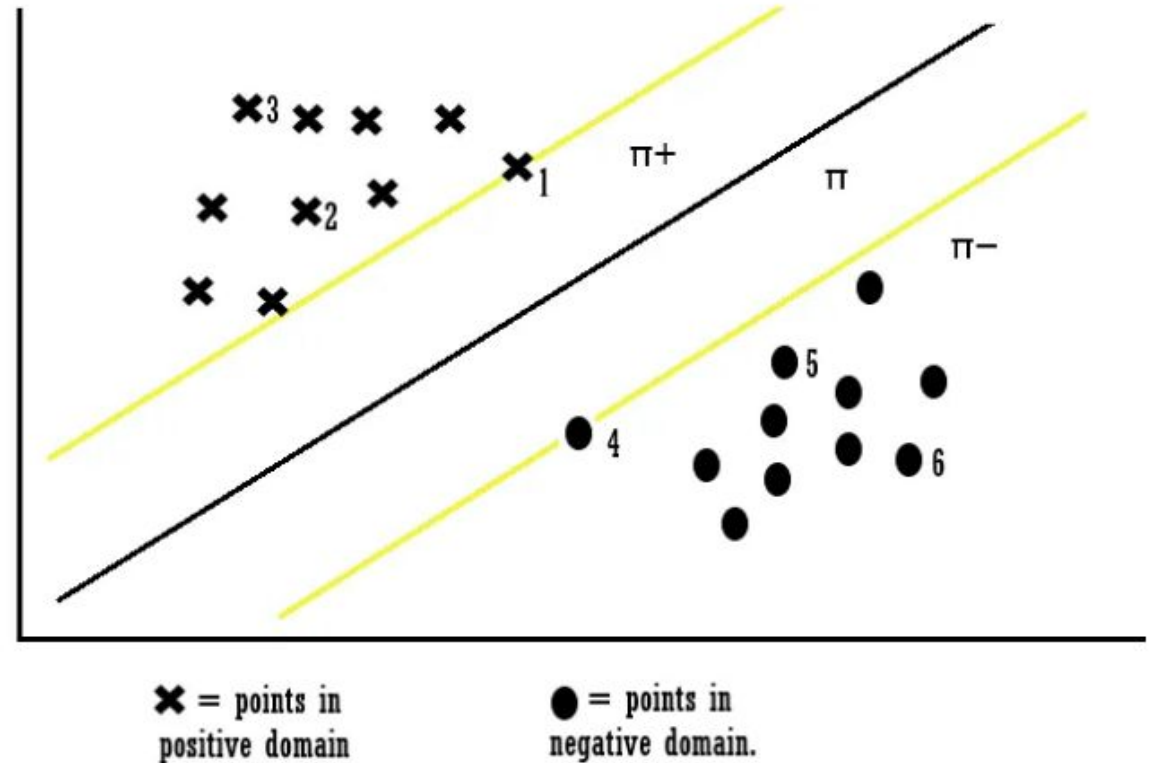
Hard Margin SVM

- If $w^T x + b \geq 1$: one class
- If $w^T x + b \leq -1$: other class
- $\pi = w^T x + b = 0$
- $\pi^+ = w^T x + b = 1$
- $\pi^- = w^T x + b = -1$



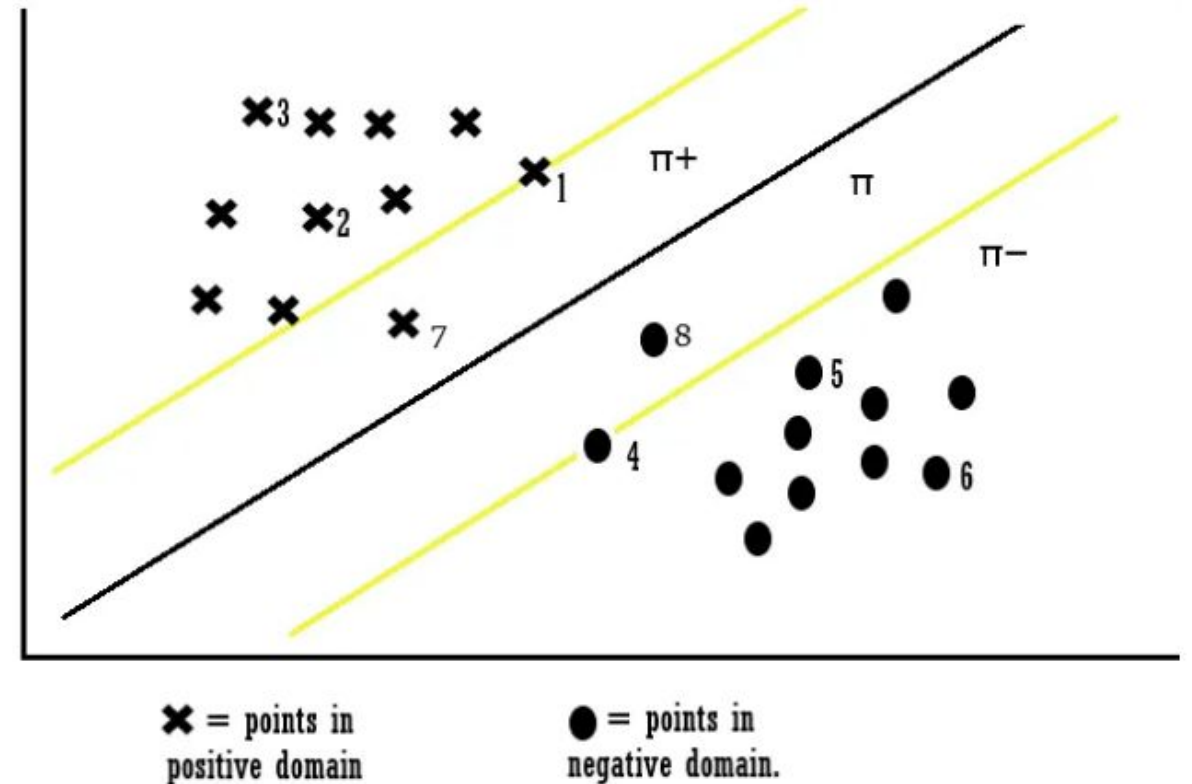
Hard Margin SVM (Cont.)

- For point x_1 ,
 $y_1 = +1$
 $w^T x_1 + b \geq 1$
- For point x_5 ,
 $y_5 = -1$
 $w^T x_5 + b \leq -1$



Hard Margin SVM (Cont.)

- If $y_i(w^T x_i + b) \geq 1$, then x_i is correctly classified
- Otherwise, x_i is incorrectly classified



Example Problem

- Given data

Obs.	X_1	X_2	Y
1	3	4	Red
2	2	2	Red
3	4	4	Red
4	1	4	Red
5	2	1	Blue
6	4	3	Blue
7	4	1	Blue

Let:

Red = +1

Blue = -1

Let $w=(w_1, w_2)$

Red (+1)

$$(3,4): 3w_1 + 4w_2 + b \geq 1$$

$$(2,2): 2w_1 + 2w_2 + b \geq 1$$

$$(4,4): 4w_1 + 4w_2 + b \geq 1$$

$$(1,4): w_1 + 4w_2 + b \geq 1$$

Blue (-1)

$$(2,1): 2w_1 + w_2 + b \leq -1$$

$$(4,3): 4w_1 + 3w_2 + b \leq -1$$

$$(4,1): 4w_1 + w_2 + b \leq -1$$

Example Problem (cont.)

At optimum, **support vectors satisfy equality**:

$$y_i(w \cdot x_i + b) = 1$$

From inspection (closest points):

Red: (2,2)

Blue: (2,1) and (4,3)

From (2,2), $y=+1$:

$$2w_1 + 2w_2 + b = 1 \quad (1)$$

From (2,1), $y=-1$:

$$2w_1 + w_2 + b = -1 \quad (2)$$

From (4,3), $y=-1$:

$$4w_1 + 3w_2 + b = -1 \quad (3)$$

Now Solve the system of linear equations (3 equations, 3 unknowns)

Subtract (2) from (1):

$$w_2 = 2$$

Now plug into (2):

$$2w_1 + b = -3 \quad (4)$$

And from (3):

$$4w_1 + b = -7 \quad (5)$$

Subtracting (4) from (5):

$$w_1 = -2$$

Plug into (4):

$$b = 1$$

Example Problem (cont.)

Final solution:

$$w = (-2, 2), b = 1$$

So, final hyperplane:

$$-2x_1 + 2x_2 + 1 = 0 \quad \text{or} \quad x_2 = x_1 - 1/2$$

Margin equations:

$$w \cdot x + b = \pm 1$$

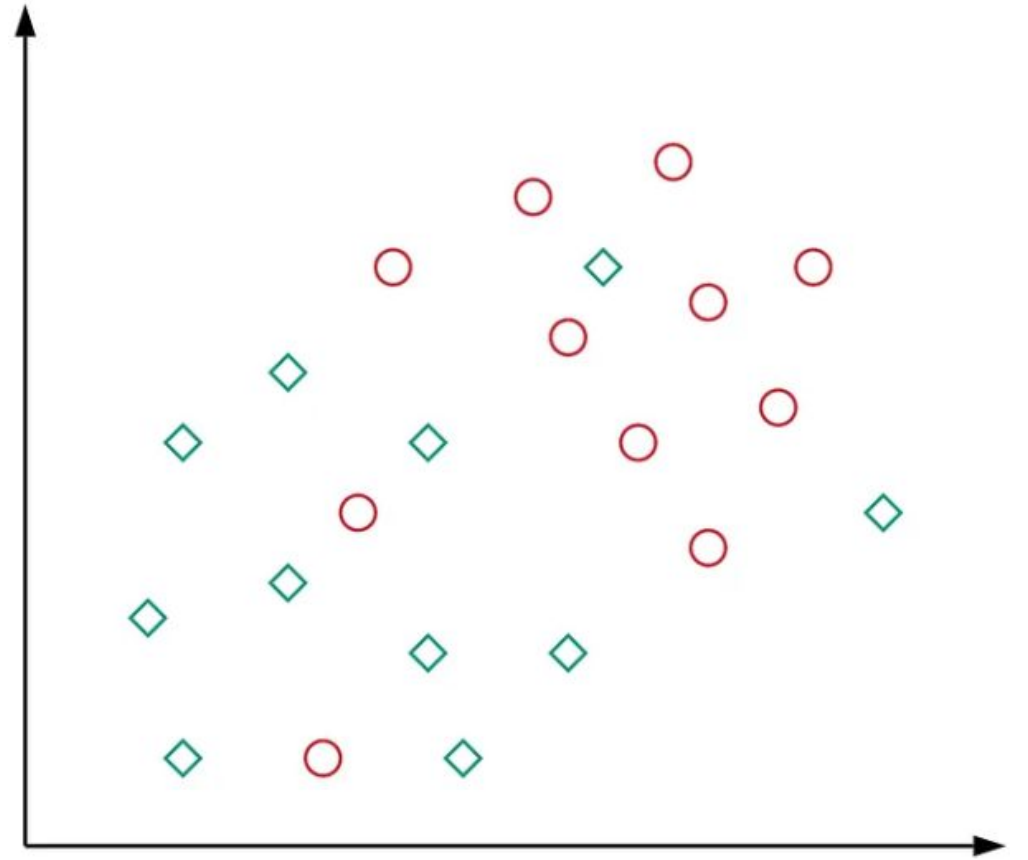
$$\begin{aligned} -2x_1 + 2x_2 + 1 &= 1 & \text{or} & \quad x_2 = x_1 \\ -2x_1 + 2x_2 + 1 &= -1 & \text{or} & \quad x_2 = x_1 - 1 \end{aligned}$$

What if...

- Data is not linearly separable
- Two solutions
 - Soft Margin SVM
 - The Kernel Trick

Linearly Inseparable Data

- A linear decision boundary cannot be drawn through this kind of dataset
- The hard and fast rule of SVM is not applicable for them

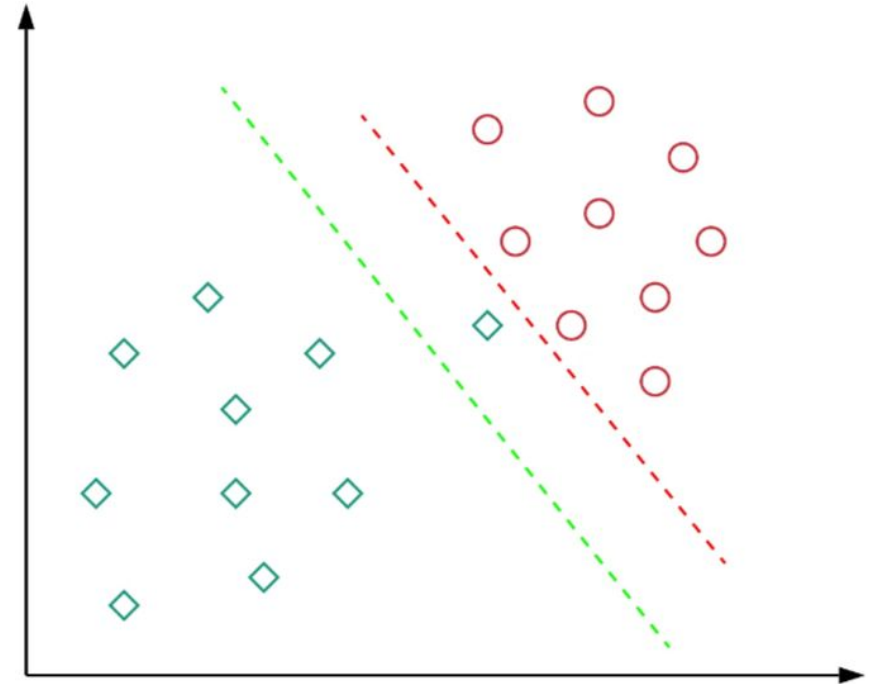


Soft Margin SVM

- Allow SVM to make a certain number of mistakes
- Keep the margin as wide as possible
- This can be done simply by modifying the objective of SVM

Benefits of Small Mistakes

- The green decision boundary can have a wider margin
- It will generalize well over unseen data points
- Also help avoid overfitting



The Mathematics

- The objective function is defined as,

$$L = \frac{1}{2} ||w||^2 + C \sum_{i=1}^n \xi_i$$

where, w is the margin width, ξ (slack variable) is how much a data point violates the margin, and C is a regularization parameter controlling trade-off between:

- margin size
- classification error (slack penalties)

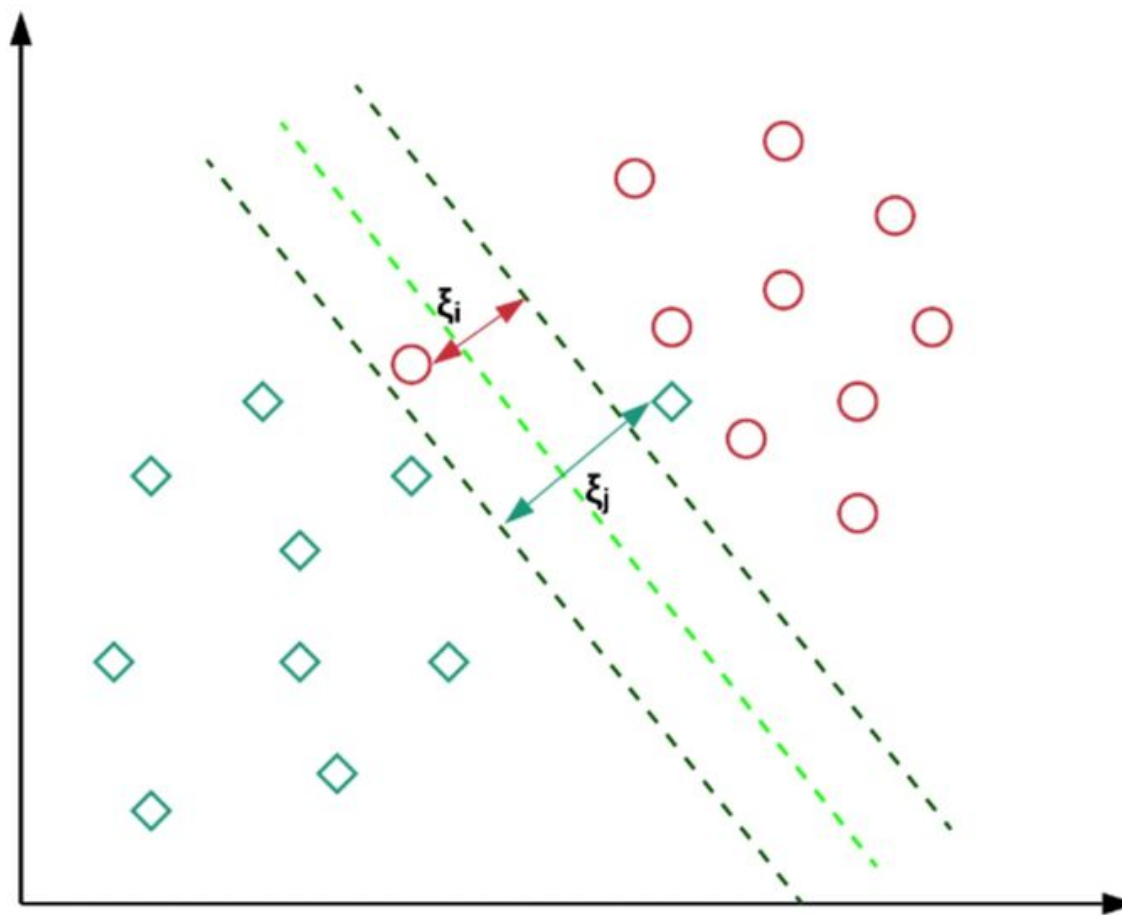
The Mathematics

- The objective function is subject to,

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$

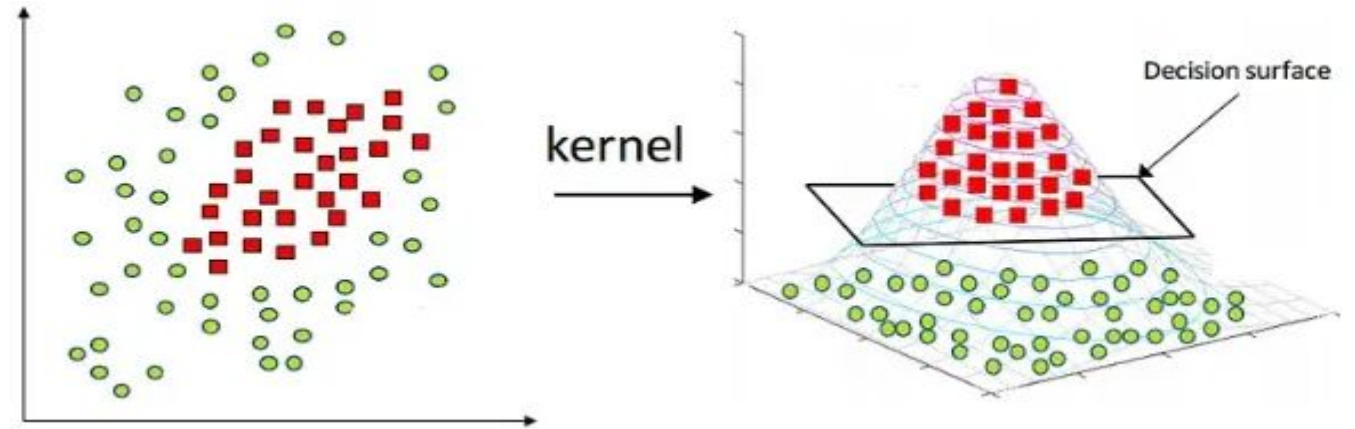
- A small **C** means focusing on maximizing **w** instead of minimizing the number of mistakes and vice versa

The Mathematics (Support)



The Kernel Trick

- Sometimes adding a dimension to the dataset makes it linearly separable



- The 2-dimensional data is not linearly separable at all
- But if we take it to 3-dimensions, then it is seen to be linearly separable

Adding a Dimension

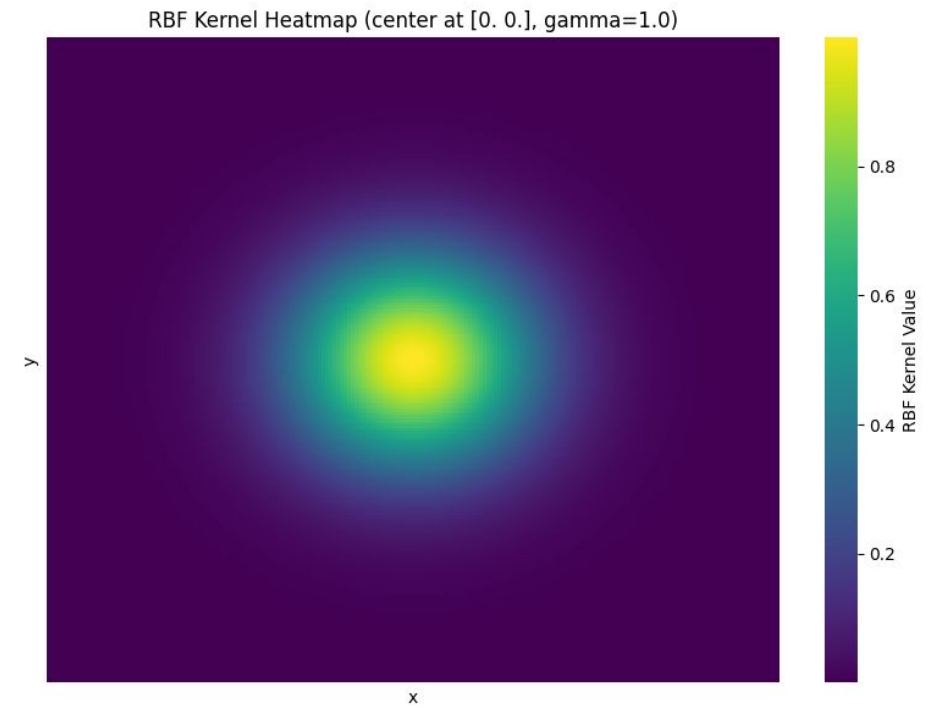
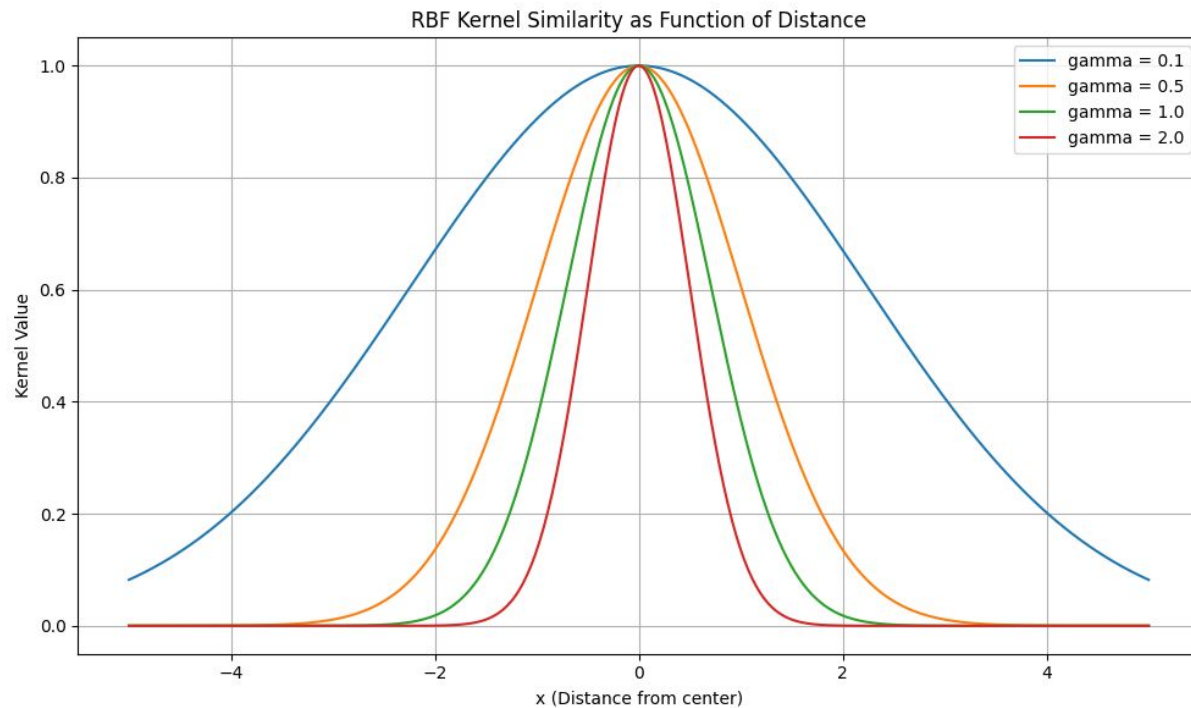
- Suppose we have a data point $(x_1, x_2, x_3, \dots, x_n)$ in n -dimensions
- We simply add a dimension to it, making it $(x_1, x_2, x_3, \dots, x_n, f(x_1, x_2, x_3, \dots, x_n))$
- Here f is the kernel function that adds a new dimension based on the data from the current dimensions

The Kernel

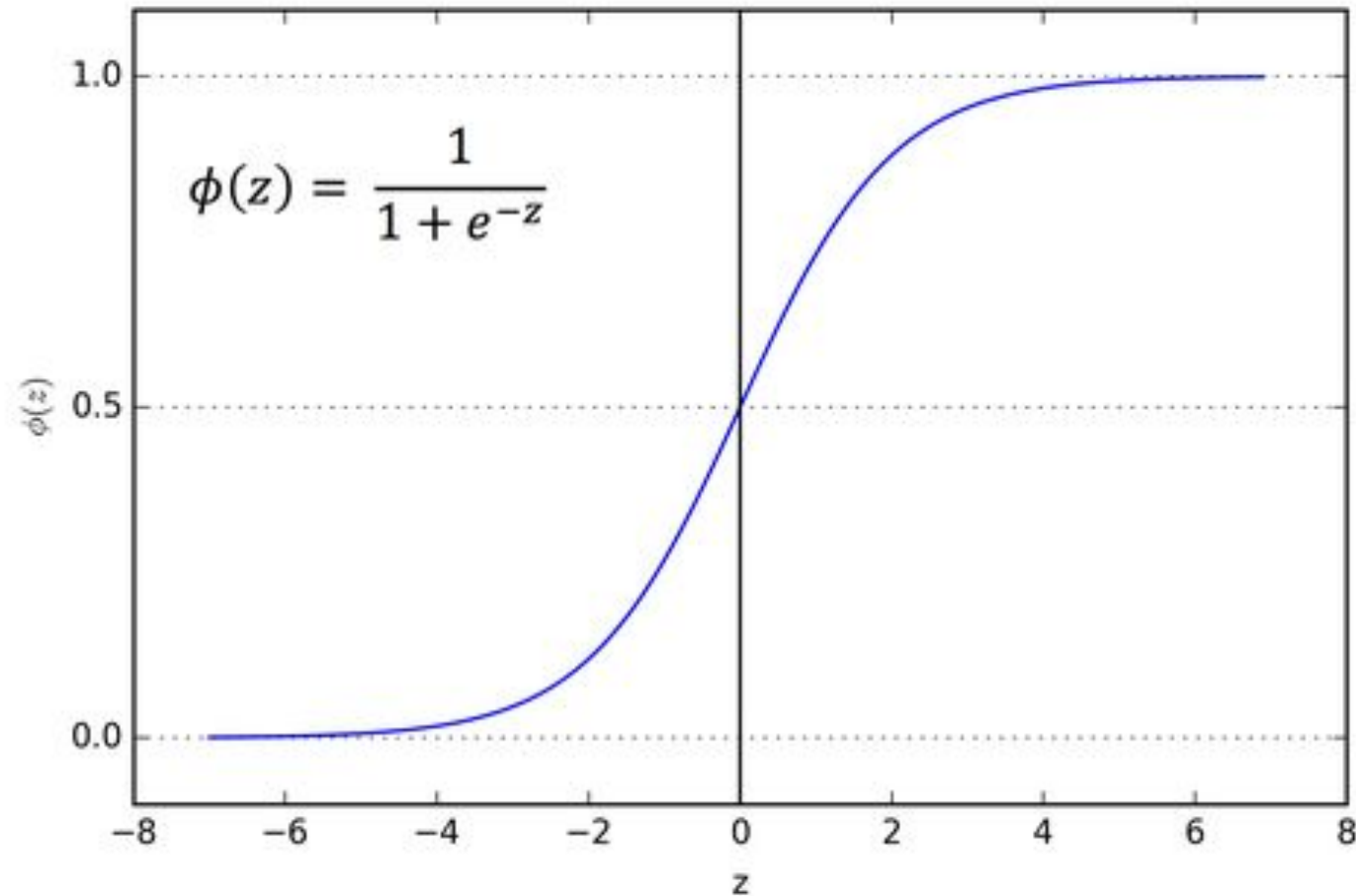
- There are several types of kernel functions
 - Gaussian kernel radial basis function (RBF)
 - Sigmoid function
 - Polynomial kernel
 - Linear kernel

Gaussian Kernel Radial Basis Function (RBF)

$$K(X_1, X_2) = e^{-\frac{||X_1 - X_2||^2}{2\sigma^2}} = e^{-\gamma ||X_1 - X_2||^2}$$



Sigmoid Function



Other Kernels

- Polynomial kernel

$$K(X, Y) = (\gamma \cdot X^T Y + r)^d, \gamma > 0$$

- Linear kernel

$$f(x_1, x_2, x_3, \dots, x_n) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \dots + w_n x_n$$

Formulation of Kernels

- The formulation of kernels may differ from books to books
- All of them are correct
- There are many more kernels available to use

Pros and Cons of SVM

- Pros

- Best for linearly separable data
- Can handle high dimensionality better, especially when the number of data points is smaller than the number of features
- Adding new data doesn't affect the model
- Memory efficient

- Cons

- Does not work well for large datasets
- Sensitive to noise and overlapping of target classes
- No probabilistic explanation for the predictions

Questions?